

Aim:- To perform SQL queries for demonstrating Insertion, Updation & deletion of records in Database tables.

Theory:- SQL provides commands to manipulate data stored in database tables. These commands are part of Data Manipulation (DML). The main DML operations are:

1. Insert :- Used to add new records into a table

Syntax INSERT INTO table-name
 VALUES (value1, value2, ...,);

2. UPDATE :- Used to modify existing records in a table.

Syntax UPDATE table-name
 SET Column-name = value
 WHERE condition;

3. DELETE :- Used to remove records from a table.

Syntax DELETE FROM table-name
 WHERE condition;

Output
Student table After Insertion

Student-ID	Name	Course	Marks
1	Rahul	B.tech	85
2	Priya	BCA	90
3	Aman	B.tech	78

Student table After Updation

Student-ID	Name	Course	Marks
1	Rahul	B.tech	85
2	Priya	B.CA	95
3	Aman	B.tech	78

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Program:-

```
CREATE DATABASE CollegeDB;
USE CollegeDB;
```

Create student table

```
CREATE TABLE student (
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(100),
    Course VARCHAR(50),
    Marks INT);
```

Insert operation

```
INSERT INTO Student VALUES
(1, 'Rahul', 'B-Tech', 85),
(2, 'Priya', 'BCA', 90),
(3, 'Aman', 'B-tech', 78);
```

```
SELECT * FROM Student;
```

Update operation-

```
UPDATE student
```

```
SET Marks = 95
```

```
WHERE Student_ID = 2;
```

```
SELECT * FROM Student
```

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Student table After Deletion

Student ID	name	Course	Marks
1	Rahul	B.tech	85
2	Priya	BCA	95

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Delete operation

DELETE RECORD FROM Student

WHERE Student ID = 3;

SELECT * FROM Student;

Result Thus, SQL operations for insertion, updation & Deletion were successfully performed on the Database table.

Viva Question

Q1 What is DML in SQL?

Ans DML (Data manipulation language) is used to manipulate data in tables.

Q2 Which command is used to insert records?

Ans INSERT INTO

Q3 Which command is used to modify existing data?

Ans UPDATE

Q4 Which command is used to remove records?

Ans DELETE

Q5 Why is WHERE clause important in UPDATE & DELETE?

Ans It specifies which records should be modified or deleted.

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Aim:- To implement Referential Integrity Constraints using SQL queries.

Theory:- Referential Integrity is a property of a relational database that maintains consistency between related tables. It ensures that:

- A foreign key value in one table must match a primary key value in another table.
 - Invalid or unmatched data cannot be inserted.
- Referential integrity is maintained using the FOREIGN KEY constraint.

Advantage of Referential Integrity

1. Maintains data consistency
2. Prevents invalid data entry
3. Maintain relationship between tables
4. Avoids orphan record.
- 5.

PRIMARY KEY Uniquely identifies each record in a table

Ex. `COURSE_ID INT PRIMARY KEY`

FOREIGN KEY - Creates relationship between tables. References primary key of another table.

Ex. `FOREIGN KEY (COURSE_ID)
REFERENCES Course (COURSE_ID)`

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Program

```
CREATE DATABASE CollegeDB;  
USE CollegeDB;
```

Create Parent table (Course)

```
GO  
CREATE TABLE Course (  
    Course_ID INT PRIMARY KEY,  
    Course_Name VARCHAR(100)  
) ;
```

Create child table (Student)

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(100),  
    Course_ID INT,  
    FOREIGN KEY (Course_ID)  
    REFERENCE Course (Course_ID)  
) ;
```

Data insertion -

```
INSERT INTO Course VALUES  
(101, 'B-tech'),  
(102, 'BCA');
```

INSERT INTO Student VALUES

```
(1, 'Rahul', 101),  
(2, 'Priya', 102);
```

```
SELECT * FROM Course;
```

OutputCourse Table

Course-ID	Course-Name
101	B.tech
102	BCA

Student Table

Student-ID	Name	Course-ID
1	Rahul	101
2	Priya	102

Error Output

Cannot add or update a child row:
a foreign key constraint fails:

SELECT * FROM Student;

INSERT INTO Student VALUES
(3, 'Aman', 105);

The above query gives an error because Course-ID
= 105 does not exist in the course table,

Result:- Thus, Referential integrity Constraints were
successfully implemented using PRIMARY KEY
& FOREIGN KEY Constraints.

 Viva Question

Q1 What is Referential Integrity?

Ans To ensure consistency between related tables in
a database.

Q2 Which constraint is used for Referential Integrity?

Ans FOREIGN KEY constraint

Q3 What is a parent table?

Ans The table containing the PRIMARY KEY

Q4 What is a child table?

Ans The table containing the FOREIGN KEY

Q5 What happens if foreign key value does not exist in
parent table?

Ans The database gives a constraint violation error

violation error

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Aim: To write SQL queries for creating users & assigning Roles / Privileges in a database.

Theory - In DBMS, multiple users can access the database.

To control database security, SQL queries provides commands to:

- Create Users
- Grant Privileges
- Revoke Privileges
- Assign Roles

A Role is a collection of permissions assigned to users.

Advantages of Roles & Privileges

1. Improves database security
2. Restrict unauthorized access
3. Simplifies permission management
4. Provides controlled data access.

Important Command:

1. CREATE USER - Used to create a new database user.

Syntax -

```
CREATE USER 'username'@'localhost'  
IDENTIFIED BY 'Password';
```

Output

User Created Successfully

User Name	Host
student 1	localhost

Granted Privileges

Privileges

SELECT

INSERT

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2. GRANT Used to assign privileges to user.

Syntax

GRANT privileges

ON Database.table

TO 'username'@'localhost';

3. REVOKE Used to remove privilege from users.

Syntax

REVOKE privilege

ON Database.table

FROM 'username'@'localhost';

Program

CREATE DATABASE CollegeDB;

USE CollegeDB;

CREATE USER 'student1'@'localhost'

IDENTIFIED BY 'student123'

GRANT SELECT

ON CollegeDB.*

TO 'student1'@'localhost';

GRANT INSERT

ON CollegeDB.*

TO 'student1'@'localhost';

SHOW GRANTS FOR 'student1'@'localhost';

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After REVOKE

Remaining Privilege

SELECT

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REVOKE INSERT

ON CollegeDB.*

FROM 'student1'@'localhost';

SHOW GRANTS FOR 'student1'@'localhost';

Result:- Thus, SQL Queries for creating users & assigning/revoking roles & privileges were successfully implemented.

Viva Question

Q1 What is a database user?

Ans A user is an account that can access the database

Q2 What is a role in DBMS?

Ans A role is a collection of privileges assigned to user

Q3 Which command is used to create a user?

Ans CREATE USER

Q4 Which command is used to assign privileges?

Ans GRANT

Q5 Which command is used to revoke privileges?

Ans REVOKE

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